

Project Planning Phase

Project Planning Template (Product Backlog, Sprint Planning, Stories, Story points)

Date	15 February 2025
Team ID	LTVIP2026TMIDS88736
Project Name	Exploratory Analysis of Rainfall Data in India for Agriculture
Maximum Marks	5 Marks

Product Backlog, Sprint Schedule, and Estimation (4 Marks)

Use the below template to create product backlog and sprint schedule

Sprint	Functional Requirement (Epic)	User Story No	User Story / Task	Story Points	Priority	Team Members
Sprint-1	User Registration	USN-1	As a user, I can register in the rainfall analysis system using email & password to access insights	2	High	Frontend, Backend
Sprint-1	User Confirmation	USN-2	As a user, I receive email confirmation after registration	1	High	Backend
Sprint-2	Social Registration	USN-3	As a user, I can register using social login for quick access	2	Low	Backend
Sprint-1	User Login	USN-4	As a user, I can login to access rainfall dashboards	1	High	Backend
Sprint-1	Dashboard	USN-5	As a farmer/researcher, I can view rainfall visualization dashboard	3	High	Frontend
Sprint-1	Data Input	USN-6	As a user, I can upload rainfall datasets for analysis	3	High	Frontend, Backend
Sprint-1	Data Preprocessing	USN-7	As a system, rainfall data is cleaned and prepared for ML models	4	High	ML Engineer
Sprint-1	ML Prediction Engine	USN-8	As a user, I can analyze rainfall trends using ML models	5	High	ML Engineer

Sprint	Functional Requirement (Epic)	User Story No	User Story / Task	Story Points	Priority	Team Members
Sprint-2	Feature Analysis	USN-9	As a user, I can view key rainfall pattern insights	3	Medium	ML Engineer
Sprint-2	Reports	USN-10	As a policymaker, I can download rainfall analysis reports	3	Medium	Frontend
Sprint-2	History	USN-11	As a user, I can view previous rainfall analyses	2	Medium	Backend
Sprint-1	Admin User Management	USN-12	Admin can manage system users	2	High	Backend
Sprint-1	Dataset Management	USN-13	Admin uploads rainfall datasets for training	3	High	ML Engineer
Sprint-1	Model Training	USN-14	Admin trains ML models for better accuracy	5	High	ML Engineer
Sprint-2	System Reports	USN-15	Admin views system analytics and usage	2	Medium	Backend

Project Tracker, Velocity & Burndown Chart: (4 Marks)

Sprint	Total Story Points	Duration	Start Date	End Date (Planned)	Completed	Release Date
Sprint-1	32	6 Days	1 Feb 2025	7 Feb 2025	32	7 Feb 2025
Sprint-2	15	6 Days	8 Feb 2025	14 Feb 2025	15	14 Feb 2025

Velocity:

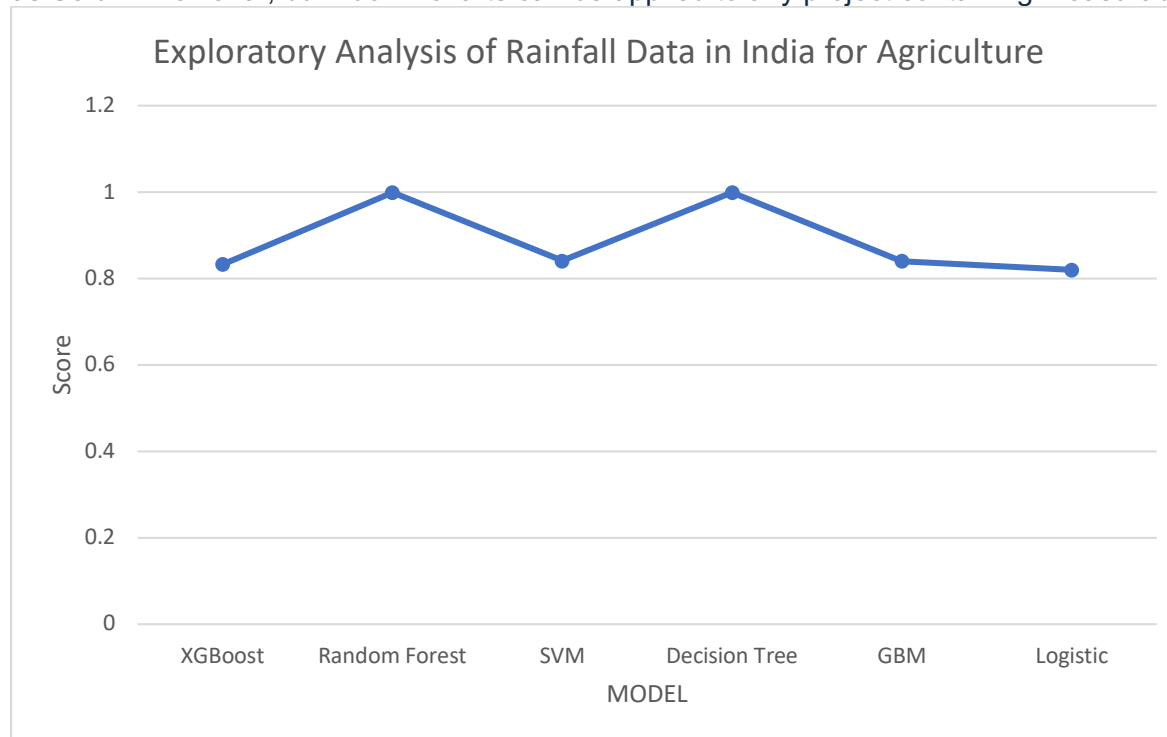
Imagine we have a 10-day sprint duration, and the velocity of the team is 20 (points per sprint). Let's calculate the team's average velocity (AV) per iteration unit (story points per day)

AV = Total Story Points / Sprint Duration

AV = 32 / 6 ≈ 5.3 story points per day

Burndown Chart:

A burn down chart is a graphical representation of work left to do versus time. It is often used in agile software development methodologies such as Scrum. However, burn down charts can be applied to any project containing measurable progress over time.



<https://www.visual-paradigm.com/scrum/scrum-burndown-chart/>

<https://www.atlassian.com/agile/tutorials/burndown-charts>